MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3450 PRESSURE OXIDATION SERVICES
DESIGN OF FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
FDN	:	Foundation

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1. INTENT

This document is to prepare design of foundation for **POX Boiler Stack (3450-STK-001).**The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel. For member strength calculation force values are directly taken from Equipment load data drawings & designed in excel sheet.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Input drawings attached in Attachment - 1. Also referred Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 & ACI 318M.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

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5.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	50
Exposed to weather or contact with ground	Column	50

5.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 325 Kn/m^2 at 3.0 m below FGL. (Refer Table 5 - BH-28 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).

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6. DESIGN CALCULATION FOR FOUNDATION

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DESIGN OF STACK FOUNDATION

	Along-X	Along-Z
Size of column :	1.60m	1.60m
Size of fdn. raft :	5.00m	5.00m
Thickness of raft :	0.80 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-3.00 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	20 kN/m^3	

SYMBOLS AND NOTATIONS

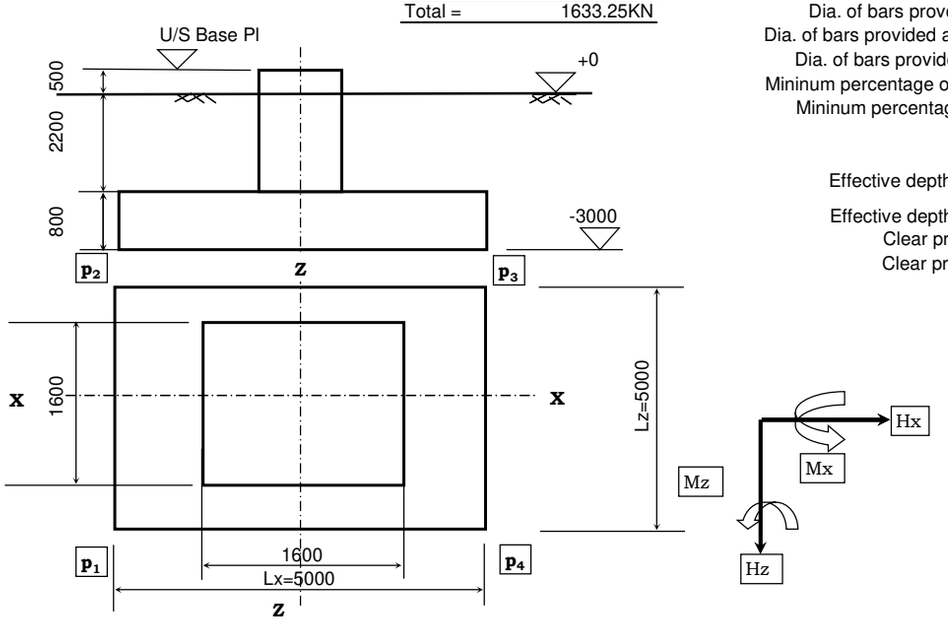
Dead Load	D
Wind Load	W
Live Load	L

Calculation of Weight of Foundation :

Weight of raft & Col.= (5*5*0.8*24)+(1.6*1.6*2.7*24) = 645.89KN

Weight of soil = ((5*5)-(1.6*1.6))*2.2*20 = 987.36KN

Total = 1633.25KN



Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	16mm
Dia. of bars provided along Z direction at bottom=	20mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 740mm = 0.74m

Effective depth of slab along Z direction = 720mm = 0.72m

Clear projection along Z direction = 1.7m

Clear projection along X direction = 1.7m

H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Dead Load (D)	51	0.0	0.0	0.0	0.0
2	Wind Load (W)	0	101	0.0	1320	0.0
3	Live Load (L)	10	0.0	0.0	0.0	0.0

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	D	51.0	0.0	0.0	0.0	0.0	1684.25	0.00	0.00	0.00	0.00
202	D + L	61.0	0.0	0.0	0.0	0.0	1694.25	0.00	0.00	0.00	0.00
203	D + 0.75 L	58.5	0.0	0.0	0.0	0.0	1691.75	0.00	0.00	0.00	0.00
204	D + W	51.0	101.0	1320.0	0.0	0.0	1684.25	101.00	1673.50	0.00	0.00
205	0.6 D + W	30.6	101.0	1320.0	0.0	0.0	1010.55	101.00	1673.50	0.00	0.00
206	D + 0.75W + 0.75L	58.5	75.8	990.0	0.0	0.0	1691.75	75.75	1255.13	0.00	0.00
Ultimate Conditions											
301	1.4 D	71.4	0.0	0.0	0.0	0.0	2357.95	0.00	0.00	0.00	0.00
302	1.2 D + 1.6 L	77.2	0.0	0.0	0.0	0.0	2037.10	0.00	0.00	0.00	0.00
303	1.2 D + 1.6 W	61.2	161.6	2112.0	0.0	0.0	2021.10	161.60	2677.60	0.00	0.00
304	0.9 D + 1.6 W	45.9	161.6	2112.0	0.0	0.0	1515.83	161.60	2677.60	0.00	0.00

Check for overturning and sliding:

	Load Cases:	Load at bottom of fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	D	1684.25	0.00	0.00	0.00	0.00	-	-	-	-
202	D + L	1694.25	0.00	0.00	0.00	0.00	-	-	-	-
203	D + 0.75 L	1691.75	0.00	0.00	0.00	0.00	-	-	-	-
204	D + W	1684.25	101.00	1673.50	0.00	0.00	2.52	-	6.67	-
205	0.6 D + W	1010.55	101.00	1673.50	0.00	0.00	1.51	-	4.00	-
206	D + 0.75W + 0.75L	1691.75	75.75	1255.13	0.00	0.00	3.37	-	8.93	-

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Load Combination & Calculation of Base pressure

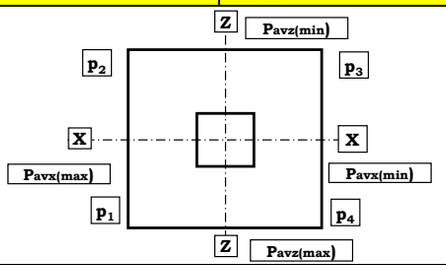
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 20 kN/m^2

Net SBC of soil for Dead Load = 325 kN/m^2 For Wind load case = 406.25 kN/m^2

Gross SBC of soil for Dead Load = 385 kN/m^2 For Wind load case = 466.25 kN/m^2



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)												Remarks
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	
201	D	1684.25	0.00	0.00	67.37	0.00	0.00	67.37	67.37	67.37	67.37	7.37	67.37	67.37	67.37	67.37	O.K
202	D + L	1694.25	0.00	0.00	67.77	0.00	0.00	67.77	67.77	67.77	67.77	7.77	67.77	67.77	67.77	67.77	O.K
203	D + 0.75 L	1691.75	0.00	0.00	67.67	0.00	0.00	67.67	67.67	67.67	67.67	7.67	67.67	67.67	67.67	67.67	O.K
204	D + W	1684.25	1673.50	0.00	67.37	80.33	0.00	147.70	147.70	-12.96	-12.96	87.70	147.70	-12.96	67.37	67.37	-
205	0.6 D + W	1010.55	1673.50	0.00	40.42	80.33	0.00	120.75	120.75	-39.91	-39.91	60.75	120.75	-39.91	40.42	40.42	-
206	D + 0.75W + 0.75L	1691.75	1255.13	0.00	67.67	60.25	0.00	127.92	127.92	7.42	7.42	67.92	127.92	7.42	67.67	67.67	O.K
301	1.4 D	2357.95	0.00	0.00	94.32	0.00	0.00	94.32	94.32	94.32	94.32	34.32	94.32	94.32	94.32	94.32	
302	1.2 D + 1.6 L	2037.10	0.00	0.00	81.48	0.00	0.00	81.48	81.48	81.48	81.48	21.48	81.48	81.48	81.48	81.48	
303	1.2 D + 1.6 W	2021.10	2677.60	0.00	80.84	128.52	0.00	209.37	209.37	-47.68	-47.68	149.37	209.37	-47.68	80.85	80.85	
304	0.9 D + 1.6 W	1515.83	2677.60	0.00	60.63	128.52	0.00	189.16	189.16	-67.89	-67.89	129.16	189.16	-67.89	60.64	60.64	

Max :	2358	2678	0
Min :	1011	0	0

Max pressure at base of foundation without surcharge excluding weight of soil (Dead load cases) = 7.37 kN/m2

Max pressure at base of foundation including surcharge excluding weight of soil (Dead load cases) = 27.37 kN/m2

Max pressure at base of foundation without surcharge excluding weight of soil (Dead+Temperature+Seismic load cases) = 149.37 kN/m2

Max pressure at base of foundation including surcharge excluding weight of soil (Dead+ Temperature+Seismic load cases) = 169.37 kN/m2

Sample calculation of values of base pressure: Load case : D + L

Maximum pressure, p_1 (pmax) = 67.77+0.00+0.00= 67.77 kN/m^2

p_2= 67.77+0.00-0.00= 67.77 kN/m^2

p_3= 67.77-(0.00+0.00)= 67.77 kN/m^2

p_4= 67.77-0.00+0.00= 67.77 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 67.77-3*20= 7.77 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (67.77+67.77)/2= 67.77 kN/m2

Minimum average Pressure along X, $p_{avx(minx)}$ = (67.77+67.77)/2= 67.77 kN/m2

Maximum average Pressure along Z, $p_{avz(max)}$ = (67.77+67.77)/2= 67.77 kN/m2

Minimum average Pressure along Z, $p_{avz(minx)}$ = (67.77+67.77)/2= 67.77 kN/m2

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Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((1.6+2*0.5*0.72)+(1.6+2*0.5*0.72))*2

= 9.28 m

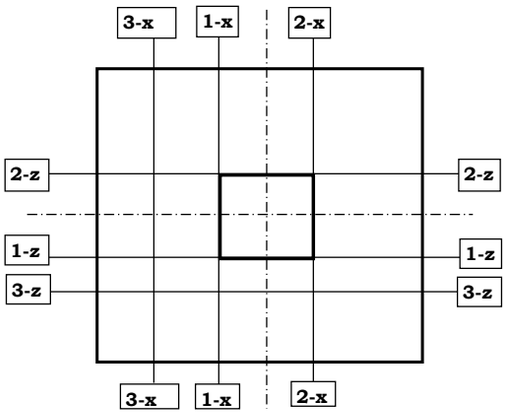
Punching Area= ((1.6+2*0.5*0.72)*(1.6+2*0.5*0.72))

= 5.38 m2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.8*24+2.2*20)

= 63.20 kN/m2

Area of footing, Af= 25.00 m2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M_{1x} kNm	B.M_{2x} kNm	S.F_{3x} kN	S.F(punching) kN
Ultimate Conditions								
301	1.4 D	94.32	94.32	94.32	44.97	44.97	29.88	610.54
302	1.2 D + 1.6 L	81.48	81.48	81.48	26.41	26.41	17.55	358.73
303	1.2 D + 1.6 W	121.97	39.72	160.02	169.12	-118.13	116.63	346.18
304	0.9 D + 1.6 W	101.76	19.51	139.81	139.92	-147.33	97.23	-50.36

Sample calculation of BM & SF

Load case : 301

p_{1x} = 94.32-(94.32-94.32)*1.7/5 = 94.32 kN/m^2

p_{2x} = 94.32+(94.32-94.32)*1.7/5 = 94.32 kN/m^2

p_{3x} = 94.32-(94.32-94.32)*(1.7-0.74)/5 = 94.32 kN/m^2

Overhang = 1.7 m

Effective depth = 0.740 m

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BM_{1x} = (94.32-63.2)*1.7*1.7/2+0.5*(94.32-94.32)*1.7*1.7*0.67 = 44.97 kN-m/m

BM_{2x} = (94.32-63.2)*1.7*1.7/2-0.5*(-94.32+94.32)*1.7*1.7*0.67 = 44.97 kN-m/m

SF_{3x} = (94.32-63.2)*(1.7-0.74)+0.5*(94.32-94.32)*(1.7-0.74) = 29.88 kN/m

$S.F_{(punching)}$ = (25-5.38)*(94.32-63.2) = 610.54 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	169.12	0.31	0.41	0.001	730.61	1332.00	20ϕT	235.86	300	150	2094.40
At top	-147.33	-0.27	0.36	0.001	635.90	444.00	16ϕT	316.19	300	150	1340.41

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi\ Vc)/bd$ = 0.740 N/mm2

Shear at a distance 'd' from face of column, Vu= 116.63 kN/m

Shear stress , τ_v= (116.63*1000)/(1000*740)

= 0.16 N/mm2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 610.54 kN

Strength of Concrete in shear Vc1 = 20313.45 kN Table 22.6.5.2_ (a)

Strength of Concrete in shear Vc2 = 12170.56 kN Table 22.6.5.2_ (b)

Strength of Concrete in shear Vc3 = 13542.30 kN Table 22.6.5.2_ (c)

Design Vc= 12170.56 kN

Allowable shear stress, $(\phi\ Vc)/bd$ = 1.33 N/mm2

Design punching shear stress, τ = (610.54*1000)/(9280*740)

= 0.09 N/mm2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p_{1z} =pressure at the face of column at section 1z-1z in kN/m2.
p_{2z} =pressure at the face of column at section 2z-2z in kN/m2.
p_{3z} =pressure at a distance d from the face of column at section 3z-3z in kN/m2.
$B.M_{1z}$ =Moment at the face of column at section 1z-1z in kN-m/m.
$B.M_{2z}$ =Moment at the face of column at section 2z-2z in kN-m/m.
$S.F_{3z}$ = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
301	1.4 D	94.32	94.32	94.32	44.97	44.97	30.50
302	1.2 D + 1.6 L	81.48	81.48	81.48	26.41	26.41	17.91
303	1.2 D + 1.6 W	80.85	80.85	80.85	25.50	25.50	17.30
304	0.9 D + 1.6 W	60.64	60.64	60.64	-3.70	-3.70	-2.51

Sample calculation of BM & SF

Load case : 301

p_{1z} =	94.32-(94.32-94.32)*1.7/5=	94.32 kN/m2		
p_{2z} =	94.32+(94.32-94.32)*1.7/5=	94.32 kN/m2	Overhang =	1.7 m
p_{3z} =	94.32-(94.32-94.32)*(1.7-0.72)/5=	94.32 kN/m2	Effective depth =	0.72 m
BM_{1z} =	(94.32-63.2)*1.7*1.7/2+0.5*(94.32-94.32)*1.7*1.7*0.67 =	44.97 kN-m/m		
BM_{2z} =	(94.32-63.2)*1.7*1.7/2-0.5*(-94.32+94.32)*1.7*1.7*0.67 =	44.97 kN-m/m		
SF_{3z} =	(94.32-63.2)*(1.7-0.72)+0.5*(94.32-94.32)*(1.7-0.72) =	30.50 kN/m		

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm2)	ρ=	Ast (mm2)	Minm. Ast (mm2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm2)
At bottom	44.97	0.087	0.116	0.0003	204.18	1296.00	20ϕT	242.41	300	150	2094.4
At top	-3.70	-0.007	-0.010	0.0000	-0.16	432.00	16ϕT	465.42	300	150	1340.4

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.74 N/mm^2

Shear at a distance 'd' from face of column, V_u= 30.50 kN/m

Shear stress , τ_v= (30.5*1000)/(1000*720)

= 0.04 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	20ϕT	150	c/c
	At Top	16ϕT	150	c/c
Along Z Direction :				
	At Bottom	20ϕT	150	c/c
	At Top	16ϕT	150	c/c

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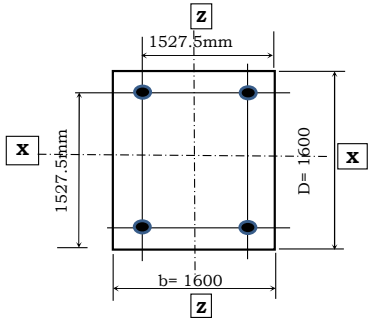
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DESIGN OF STACK COLUMN

Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1 of SBC-304
Dia. of main reinf.=	25mm	
Dia. of link=	10mm	
Effective cover =	72.5mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	1.00%	
Max. Steel Ratio	8.00%	



For Compression-controlled sections

Strength reduction factor ϕ= 0.65 Table 21.2.2 of SBC-304

Unsupproted length of column about X (l_{ux}) =	2700mm
Unsupproted length of column about Z (l_{uz}) =	2700mm

For Columns not braced against sidesway, Where, Effective length factor K= 2

Slenderness ratio,	K l_{ux} / r_x =	11.25	< 22 SECTION 6.2.5a	Radius of gyration r_x=	480	mm
	K l_{uz} / r_z =	11.25	< 22 SECTION 6.2.5a	Radius of gyration r_z=	480	mm

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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Dead Load (D)	51	0.0	0.0	0.0	0.0
2	Wind Load (W)	0	101	0.0	1320	0.0
3	Live Load (L)	10	0.0	0.0	0.0	0.0

γ_x =	0.92
γ_z =	0.92

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 D	30.36	0.0	0.0	46.7	0.0	0.0	0.01
302	1.2 D + 1.6 L	42.03	0.0	0.0	64.7	0.0	0.0	0.01
303	1.2 D + 1.6 W	26.03	254.8	0.0	40.0	392.0	0.0	0.00
304	0.9 D + 1.6 W	19.52	254.8	0.0	30.0	392.0	0.0	0.00

Reinforcement: Provide,

4-Nos	25 mm Dia	1964.29 Sq mm (At Corner)	Asc pvd. = 17678.57 Sq mm
16-Nos	25 mm Dia	7857.14 Sq mm (Along Z axis)	
16-Nos	25 mm Dia	7857.14 Sq mm (Along X axis)	

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 8305.9063 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.9		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
301	0.05	0.05	716.8	716.8	0.55	1.16	0.00	O.K
302	0.05	0.05	716.8	716.8	0.55	1.16	0.00	O.K
303	0.05	0.05	716.8	716.8	0.55	1.16	0.50	O.K
304	0.05	0.05	716.8	716.8	0.55	1.16	0.50	O.K

Links, 10mm dia @ 300 C/C

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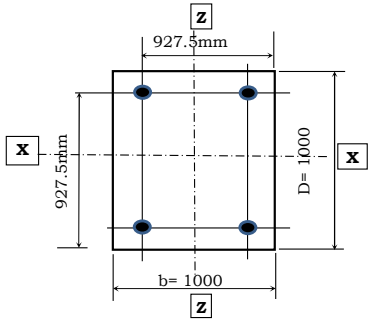
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As the column section is oversafe , trial section of 1000mm x 1000mm is used for strength and reinforecement calculation as per clause no 10.3.1.2 of SBC-304.

DESIGN OF STACK COLUMN

Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1 of SBC-304
Dia. of main reinf.=	25mm	
Dia. of link=	10mm	
Effective cover =	72.5mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	1.00%	
Max. Steel Ratio	8.00%	



For Compression-controlled sections

Strength reduction factor ϕ= 0.65 Table 21.2.2 of SBC-304

Unsupproted length of column about X (l_{ux}) =	2700mm
Unsupproted length of column about Z (l_{uz}) =	2700mm

For Columns not braced against sidesway, Where, Effective length factor K= 2

Slenderness ratio,	$K l_{ux} / r_x$ =	18.00	< 22 SECTION 6.2.5a	Radius of gyration r_x=	300	mm
	$K l_{uz} / r_z$ =	18.00	< 22 SECTION 6.2.5a	Radius of gyration r_z=	300	mm

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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Dead Load (D)	51	0.0	0.0	0.0	0.0
2	Wind Load (W)	0	101	0.0	1320	0.0
3	Live Load (L)	10	0.0	0.0	0.0	0.0

γ_x = 0.88

γ_z = 0.88

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 D	16.21	0.0	0.0	24.9	0.0	0.0	0.00
302	1.2 D + 1.6 L	31.19	0.0	0.0	48.0	0.0	0.0	0.01
303	1.2 D + 1.6W	13.90	254.8	0.0	21.4	392.0	0.0	0.00
304	0.9 D + 1.6W	10.42	254.8	0.0	16.0	392.0	0.0	0.00

Reinforcement: Provide,

4-Nos	25 mm Dia	1964.29 Sq mm (At Corner)	Asc pvd. = 17678.57 Sq mm
16-Nos	25 mm Dia	7857.14 Sq mm (Along Z axis)	
16-Nos	25 mm Dia	7857.14 Sq mm (Along X axis)	Total Asc = 1.77 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 3664.9063 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.9		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_c b^2 D$	$R_{nx}=M_{nox}/f'_c b^2 D$						
301	0.05	0.05	716.8	716.8	0.55	1.16	0.00	O.K
302	0.05	0.05	716.8	716.8	0.55	1.16	0.00	O.K
303	0.05	0.05	716.8	716.8	0.55	1.16	0.50	O.K
304	0.05	0.05	716.8	716.8	0.55	1.16	0.50	O.K

Links, 10mm dia @ 300 C/C

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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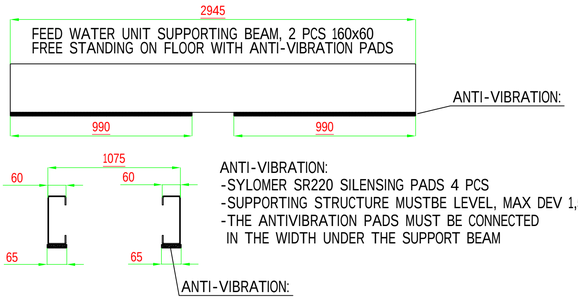
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ATTACHMENT 1 INPUT DRAWINGS

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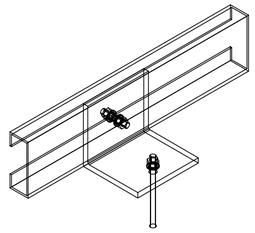
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DRAWING NUMBER MD-216-3450-EG-ME-GAD-1014 REV D01

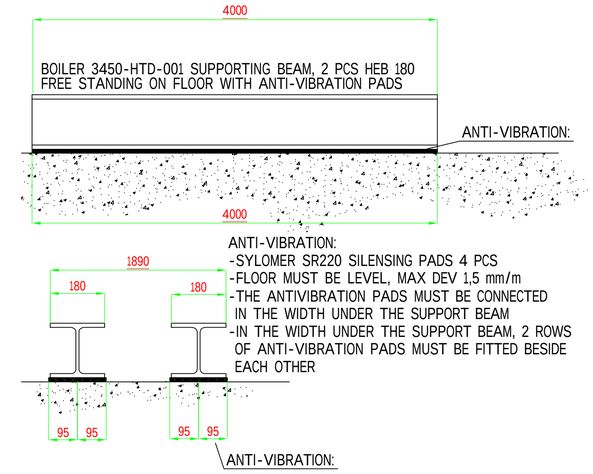


ANTI-VIBRATION:
-SYLOMER SR220 SILENSING PADS 4 PCS
-SUPPORTING STRUCTURE MUSTBE LEVEL, MAX DEV 1,5 mm/m
-THE ANTIVIBRATION PADS MUST BE CONNECTED IN THE WIDTH UNDER THE SUPPORT BEAM

P2: FEED WATER UNIT, SUPPORTING STRUCTURE

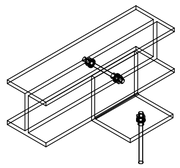


FIXING FEED WATER TANK 3450-TNK-002 FOR SEISMIC MOVEMENT
-L-STEEL 180*180*15 l=200
-4PCS.
-THREADED ROD M16 L=ACC TO STRUCTURAL DESIGN, 4 PCS
-THREADED ROD M16 l=80, 4 PCS
-NUT+WASHER M16 18 PCS

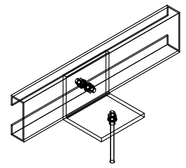


ANTI-VIBRATION:
-SYLOMER SR220 SILENSING PADS 4 PCS
-FLOOR MUST BE LEVEL, MAX DEV 1,5 mm/m
-THE ANTIVIBRATION PADS MUST BE CONNECTED IN THE WIDTH UNDER THE SUPPORT BEAM
-IN THE WIDTH UNDER THE SUPPORT BEAM, 2 ROWS OF ANTI-VIBRATION PADS MUST BE FITTED BESIDE EACH OTHER

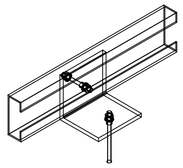
P1: BOILER SUPPORTING STRUCTURE 2 PCS



P1: FIXING BOILER 3450-HTD-001 FOR SEISMIC MOVEMENT
-L-STEEL 180*180*15 l=200
-3+3 PCS.
-EXPANSION ANCHOR M16 6PCS
-THREADED ROD M16 l=180 6PCS
-NUT+WASHER M16 18 PCS



P3,P4: FIXING PUMPS 3450-PMP-007...010 FOR SEISMIC MOVEMENT
-L-STEEL 180*180*15 l=200
-4 PCS
-EXPANSION ANCHOR M16 6 PCS
-THREADED ROD M16 l=80 4 PCS
-NUT+WASHER M16 12 PCS

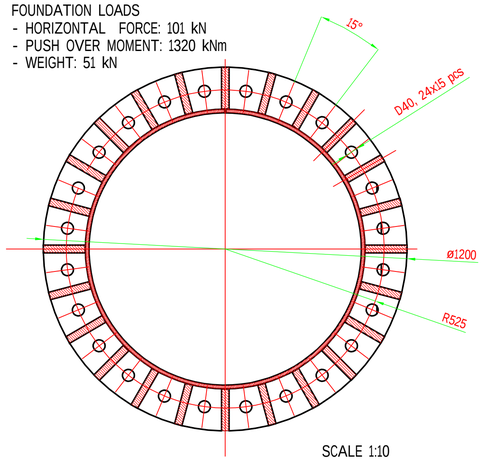


P7: FIXING BEM 3450-TNK-003 FOR SEISMIC MOVEMENT
-L-STEEL 180*180*15 l=200
-2 PCS.
-EXPANSION ANCHOR M16 2 PCS
-THREADED ROD M16 l=130 2 PCS
-NUT+WASHER M16 6 PCS

P8: STACK SUPPORTING STRUCTURE

P8 STACK, FOUNDATION BOLTS:
-24 PCS, M33
-GREATEST TENSION LOAD: 201 kN

FOUNDATION LOADS
- HORIZONTAL FORCE: 101 kN
- PUSH OVER MOMENT: 1320 kNm
- WEIGHT: 51 kN



SCALE 1:10

0 1 10m

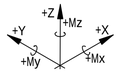
GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

DIRECTION OF FORCES AND MOMENTS



05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
LP = LOW POINT
HP = HIGHT POINT

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
OPERATING PLATFORM- GRATING 5 kN/m² GROUNDFLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²
HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1,6 FOR STRENGTH DESIGN.

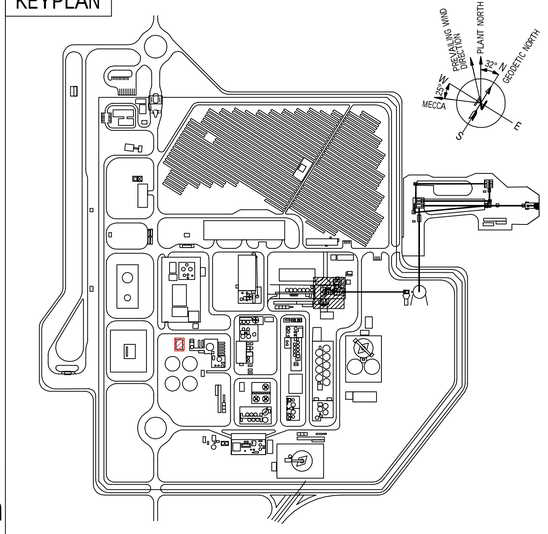
13 - REPORTED SEISMIC LOAD ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301-2007. HOWEVER, IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNER.

14 - FINISHED GROUND LEVEL (F.G.L.) = 912.850 m.

15 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

16 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



LOADING POINT		P1			P2			P3			P4			P5			P6			P7			P8				P9			P10		
	LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	M[kNm]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]
1	DEAD LOAD EQUIPMENT	– 300			– 20			– 4			– 4			– 4			– 1			– 3			– 51			– 1			– 1		
2	DEAD LOAD (SECONDARY EQP)	– 10			– 10																										
3	OPERATION LOAD 1. FILLING	– 400			– 60			– 5			– 5			– 6			– 2			– 5						– 2			– 2		
4	OPERATION LOAD 2. PRESSURE																														
5	OPERATION LOAD 3. TEMPERATURE																														
6	OPERATION LOAD 4. DYNAMIC							– 6			– 6																				
7	OPERATION LOAD 5. OTHERS																														
8	TEST LOAD FILLING / EMERGENCY				– 72															– 8											
9	LIVE LOAD																														
10	WIND +- X - DIRECTION																														
11	WIND +- Y - DIRECTION																														
12	EARTHQUAKE +- X - DIRECTION			20	20		3	3		0,25	0,25		0,25	0,25		0,3	0,3		0,1	0,1		0,1	0,1					0,1	0,1		0,1	0,1
13	EARTHQUAKE +- Y - DIRECTION			20	20		3	3		0,25	0,25		0,25	0,25		0,3	0,3		0,1	0,1		0,1	0,1					0,1	0,1		0,1	0,1
14	HORIZONTAL FORCE																							101								
15	PUSH OVER MOMENT																								1320							
	EQUIPMENT - NO.	3450-HTD-001			3450-TNK-002			3450-PMP-008/010			3450-PMP-007/009			NA			3450-PKG-002			3450-TNK-003			3450-STK-001				3450-PMP-011			3450-PMP-012		
	DESCRIPTION	BOILER			FEED WATER TANK			BOOSTER PUMP/ FEED WATER PUMP			BOOSTER PUMP/ FEED WATER PUMP			CONTROL PANEL			GAS MODULE			BLOW-DOWN EXPANSION AND COOLING TANK			POX BOILER STACK				CHEMICAL DOSING PUMP			CHEMICAL DOSING PUMP		

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	21.11.2023	AS BUILT	RW	TV	SC	RL	
MD-216-3450-EG-ME-GAD-1012	LOAD PLAN, PLAN AT EL. +913.350	C01	22-12-2020	APPROVED FOR CONSTRUCTION	RW	TV	SC	RL	
MD-216-3450-EG-ME-GAD-1013	LOAD PLAN, PLAN AT EL. +916.050	A01	03-08-2020	ISSUED FOR APPROVAL	MK	TV	XK	SC	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

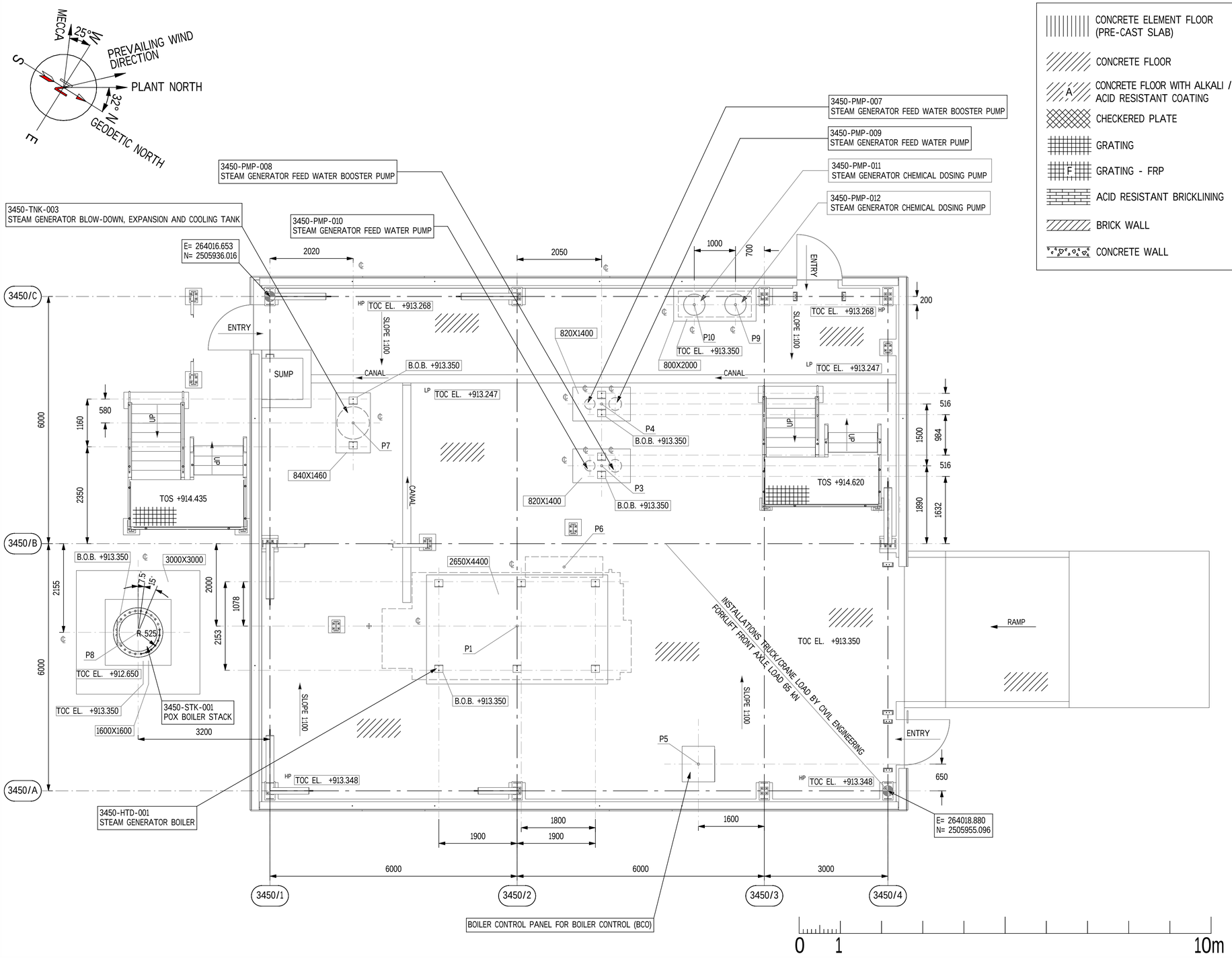
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LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_450_U_D_23149_007			2
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707 _GAU01 _HH12 _MLD14 _34505		OU100694148	2
DRAWN	MARKAJ	31-07-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TATVIE	31-07-2020	LOAD PLAN
ENGINEER	XTKA	31-07-2020	PRESSURE OXIDATION - SERVICES
APPROVED	SOUCHA	31-07-2020	FOUNDATION DETAILS
EMPLOYER			
SCALE 1:50		A1	DRAWING NUMBER: MD-216-3450-EG-ME-GAD-1014 REV: D01

DRAWING NUMBER MD-216-3450-EQ-ME-GAD-1006 REV A03



- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- A — CONCRETE FLOOR WITH ALKALI / ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- F — GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
LP = LOW POINT
HP = HIGHT POINT

DIRECTION OF FORCES AND MOMENTS: +Z, +Mz, +X, +Mx, +Y, +My

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
OPERATING PLATFORM-GRATING 5 kN/m² — GROUNDFLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²
HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1,6 FOR STRENGTH DESIGN.

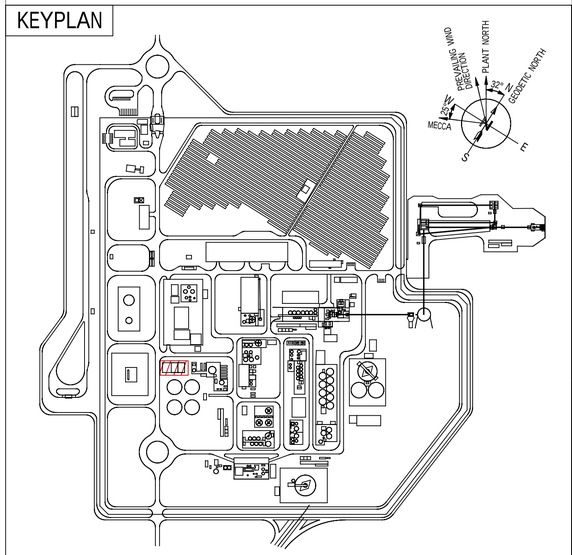
13 - REPORTED SEISMIC LOAD ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301-2007. HOWEVER, IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNER.

14 - FINISHED GROUND LEVEL (F.G.L.) = m.

15 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

16 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A03	09.12.2020	ISSUED FOR DESIGN	MKA	HKA	XTKA	XHNE	
MD-216-3450-EG-ME-GEA-1014	FOUNDATION DETAILS	A02	31.07.2020	UPDATED, ISSUED FOR DESIGN	XTKA	HKA	XTKA	XHNE	
MD-216-3450-EG-ME-GEA-1007	STEAM GEN. BOILER HOUSE LOADPLAN +916.050	A01	17.07.2020	ISSUED FOR DESIGN	XTKA	HKA	XTKA	XHNE	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045_080_U_D_23060_001 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707 _GAU01 _HH12 _MTB02 _34501 — DOCUMENT ID: 100690695 — REVISION: 2

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DRAWN	XTKA	17.07.2020
CHECKED	HKA	17.07.2020
ENGINEER	XTKA	17.07.2020
APPROVED	HNE	17.07.2020
EMPLOYER		
SCALE	1:50	A1

MANSOURAH-MASSARAH GOLD PROJECT
LOADPLAN
PRESSIRE OXIDATION -SERVICES
STEAM GEN. BOILER HOUSE +913.150

DRAWING NUMBER: MD-216-3450-EQ-ME-GAD-1006 — REV: A03